

$$h_{11} = h_i = \frac{V_1}{I_1} \Rightarrow \text{Input resistance} \dots \dots \dots (3)$$

$$h_{12} = h_r = \frac{V_1}{V_2} \Rightarrow \text{Reverse transfer voltage gain} \dots \dots \dots (4)$$

$$h_{21} = h_f = \frac{I_2}{I_1} \Rightarrow \text{Forward transfer current gain} \dots \dots \dots (5)$$

$$h_{22} = h_o = \frac{I_2}{V_2} \Rightarrow \text{Output admittance} \dots \dots \dots (6)$$

1.21 Equivalent Circuit using H-parameters

Fig. 4(a) shows a two-port linear circuit. The h-parameter equivalent circuit of a two-port linear circuit is drawn using the h-parameter Equations (1) and (2).

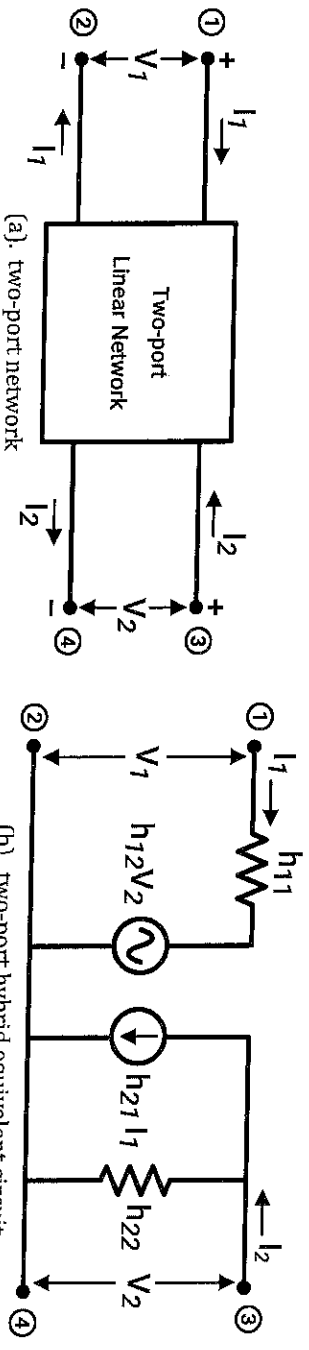


Fig.4. Hybrid equivalent circuit of a two-port system

$$V_1 = h_{11} I_1 + h_{12} V_2 \dots \dots \dots (1)$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \dots \dots \dots (2)$$

Fig.4(b) shows the h-parameter equivalent circuit of Fig.4(a) and it is derived from Eqns (1) and (2) as follows. Since the unit of measurement for each term of Eqn (1) is volts, Kirchhoff's Voltage Law (KVL) is applied to obtain the series input circuit indicated in Fig.4(b). Also, the unit of measurement of each term of Eqn (2) has the units of current; Kirchhoff's Current Law (KCL) is applied resulting in the parallel elements of the output circuit in Fig.4(b).

Note that the input circuit has a voltage-controlled voltage source whose controlling voltage is the output terminal voltage, while the output circuit has a current-controlled current source whose controlling current is the current of the input circuit. Also, the input circuit is a Thevenin's equivalent circuit while the output is Norton's equivalent circuit.

1.22 Equivalent Circuits for Transistors using h-parameters

H-parameters developed for two-port networks can also used to describe the voltage and current relationships for all three transistor circuit configurations; CE, CB and CC.

Different subscripts are used for each circuit configuration. When using h-parameters for transistor circuits, a second subscript is added to the parameters in Eqns (3), (4), (5), (6) to show the particular configuration. Therefore, the h-parameter notation for the three transistor configurations is shown in Table 1.

Table 1. H-parameter notation for transistor configurations

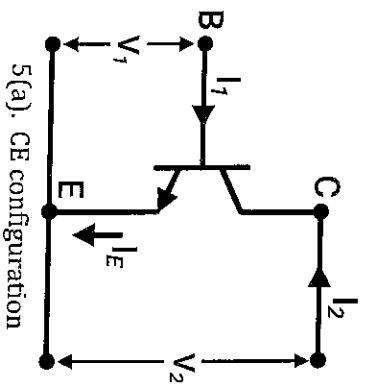
h-parameter	CE	CB	CC
h_i	h_{ie}	h_{ib}	h_{ic}
h_r	h_{re}	h_{rb}	h_{rc}
h_f	h_{fe}	h_{fb}	h_{fc}
h_o	h_{oe}	h_{ob}	h_{oc}

Common Emitter Transistor Configuration

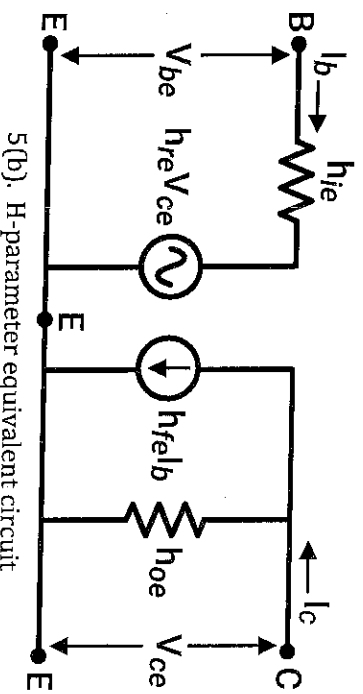
Fig. 5(a) shows an NPN transistor connected in CE configuration. The equivalent circuit for this configuration using h-parameters is shown in Fig. 5(b). The V-I relationship are described by Eqns (7) and (8).

$$V_{be} = h_{ie} I_b + h_{ce} V_{ce} \dots\dots\dots(7)$$

$$I_c = h_{fe} I_b + h_{oe} V_{ce} \dots\dots\dots(8)$$



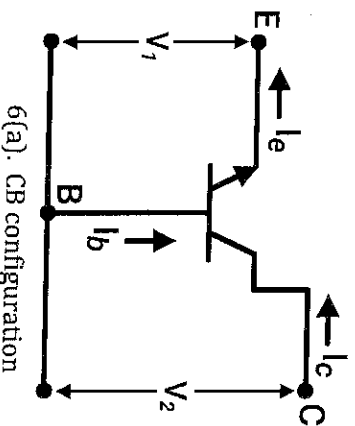
5(a). CE configuration



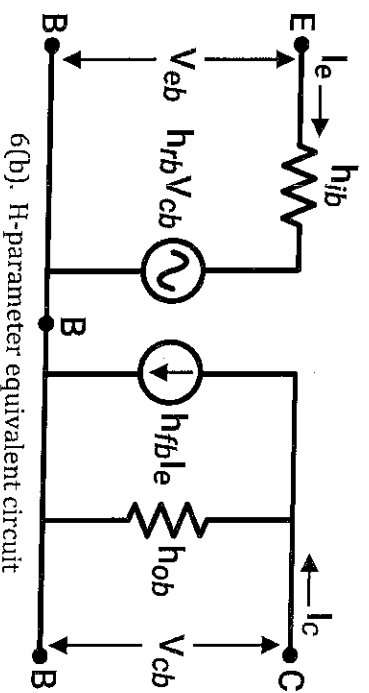
5(b). H-parameter equivalent circuit

Common Base Transistor Configuration

Fig. 6(a) shows an NPN transistor connected in CB configuration. The equivalent circuit for this configuration using h-parameters is shown in Fig. 6(b). The V-I relationship are described by Eqns (9) and (10).



6(a). CB configuration



6(b). H-parameter equivalent circuit

$$V_{eb} = h_{ib} I_e + h_{rb} V_{cb} \dots\dots\dots(9)$$

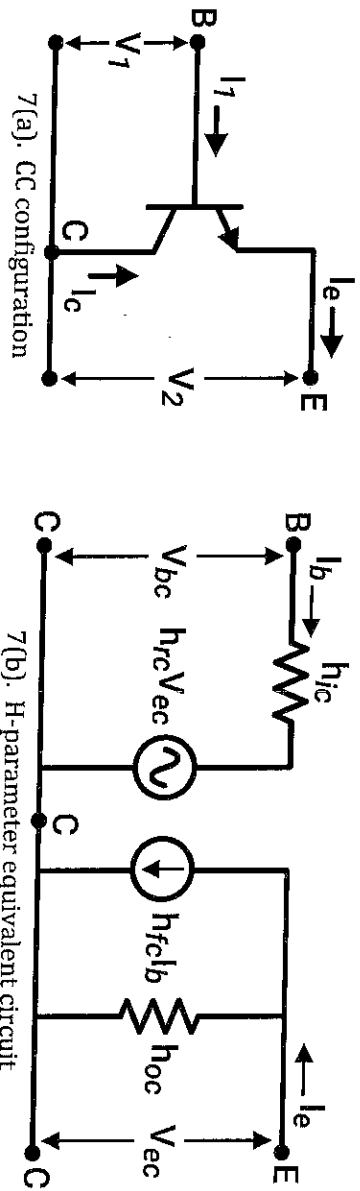
$$I_c = h_{fb} I_e + h_{ob} V_{cb} \dots\dots\dots(10)$$

Common Collector Transistor Configuration

Fig. 7(a) shows an NPN transistor connected in CC configuration. The equivalent circuit for this configuration using h-parameters is shown in Fig. 7(b). The V-I relationship are described by Eqns (11) and (12).

$$V_{bc} = h_{ic} I_b + h_{rc} V_{ec} \dots\dots\dots(9)$$

$$I_e = h_{fc} I_b + h_{oc} V_{ec} \dots\dots\dots(10)$$



1.23 Expressions for Transistor Amplifier Analysis using h-parameters

There are several circuit features in terms of h-parameters shared by all transistors whatever the configuration that are used for the analysis of transistor amplifiers. They are as follows:

- (1). Current gain (2). Input resistance (3). Voltage gain
- (4). Output impedance (5). Power gain

The expressions for these circuit features can be derived in a general way for all the three configurations of a transistor. The parameters can then be adopted for individual configurations by changing the second subscript on the h-parameters. Just like equations can be formed from circuits, circuits can be formed from equations.

Considering the general form of h-parameter Equation in (11) and (12) below,

$$\text{Input voltage, } V_{in} = h_i I_{in} + h_r V_{out} \dots\dots\dots(11)$$

$$\text{Output current, } I_{out} = h_f I_{in} + h_o V_{out} \dots\dots\dots(12)$$

Fig.8 shows the h-parameter equivalent circuit representing Eqns. (11) and (12). This h-parameter equivalent model is useful for small-signal, low-frequency operation only because, **for small input signals, a transistor behaves as a linear circuit.** **Therefore, its operation can be analyzed using h-parameters.** If the input

and the external capacitors will affect the behaviour of the transistor.

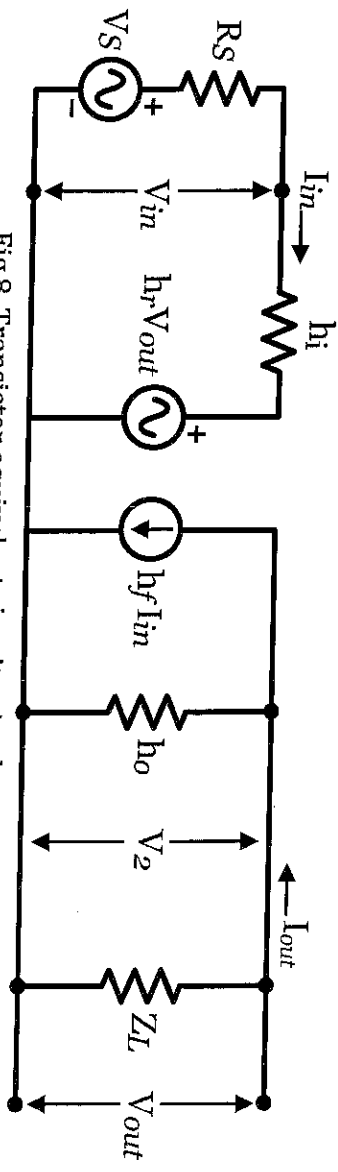


Fig.8. Transistor equivalent circuit using h-parameters
(for small-signal, low-frequency amplifiers)

(1) Current Gain, A_I

When Z_L is connected across the output port in Fig. 8, current I_L flows through Z_L .

Output voltage, $V_{out} = I_L Z_L = -I_{out} Z_L$

Current gain, $A_I = \frac{I_L}{I_{in}} = \frac{-I_{out}}{I_{in}}$

From Fig. 8 above,

$$I_{out} = h_f I_{in} + h_o (I_{out} Z_L)$$

$$\therefore I_{out} (1 + h_o Z_L) = h_f I_{in}$$

$$\frac{I_{out}}{I_{in}} = \frac{h_f}{(1 + h_o Z_L)}$$

$$A_I = \frac{I_L}{I_{in}} = \frac{-I_{out}}{I_{in}} = \frac{-h_f}{(1 + h_o Z_L)}$$

$$\text{Current Gain, } A_I = \frac{-I_{out}}{I_{in}} = \frac{-h_f}{(1 + h_o Z_L)}$$

(2) Input Impedance (Resistance), Z_i

From Eqn. (11), $V_{in} = h_i I_{in} + h_r V_{out}$

$$\therefore Z_{in} = \frac{V_{in}}{I_{in}} = h_i + \frac{h_r V_{out}}{I_{in}}$$

But, $V_{out} = I_{out} Z_L$ substituting it into the equation above,

$$Z_{in} = h_i + \frac{h_r}{1} \cdot \frac{I_{out}}{I_{in}} \cdot \frac{Z_L}{1}$$

$$\text{Substituting } A_I = \frac{-I_{out}}{I_{in}} = \frac{-h_f}{(1 + h_o Z_L)} \text{ above, we get}$$

Input impedance, $Z_{in} = h_i - \frac{h_f \cdot h_r \cdot Z_L}{(1 + h_o Z_L)}$

(3) Voltage Gain, A_V

By definition, the voltage gain of an amplifier is defined as, $A_V = \frac{V_{out}}{V_{in}}$

But, $V_{out} = I_{out} Z_L$ and $V_{in} = -I_{out} Z_{in}$

Substituting these values into the expression above we have,

$$A_V = \frac{-I_{out} Z_L}{I_{in} Z_{in}} = A_I \cdot \frac{Z_L}{Z_{in}}$$

Voltage Gain, $A_V = A_I \cdot \frac{Z_L}{Z_{in}}$

(4) Output Impedance, Z_o

The output impedance of an amplifier is defined by, $Z_o = \left. \frac{V_o}{I_o} \right|_{V_s=0}$ Also, $Z_o = \frac{1}{Y_o}$

$$I_{out} = h_f I_{in} + h_o V_{out}$$

$$\therefore Y_o = \frac{I_{out}}{V_{out}} = h_f \cdot \frac{I_{in}}{V_{out}} + h_o \quad \text{with } V_s = 0 \dots\dots\dots (13)$$

From Fig. 8, with $V_s = 0$,